

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-004-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-006-1**

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**
WPS No. **RNC-WPS-006-1** Date **9-Jun-12** Revision No. **0**
Supporting PQR No.(s) **RNC-PQR-006-1** Date **9-Jun-12** Type(s) **Manual**
Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**
(GTAW+SMAW)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1,2** to P-No. **1** Group No. **1,2**

Material Specification Type and Grade **SA 333 Gr.1,6 SA420 WPL6** TO **SA 333 Gr.1,6 SA420WPL6**

Chem. Analysis and Mech. Prop. _____

Thickness Range :

Base Metal : Groove **1.6 mm - 14.22 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other _____

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.28	5.5
AWS No. (Class)	ER-70S-G	E8016-G
F-No.	6	4
A-No.	1	1
Size of Filler Metals	Ø 2.0 & 2.4 mm	Ø 3.2 mm.
Deposited Weld Metals	2.0 mm	5.11 mm.
Thickness Range :		
Groove	4 mm Max.	10.22 mm. Max.
Fillet	All	All
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco TGS- 1N or Equivalents	Kobelco NB- 1 or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**
Welding Progression Up ☒ X Down ☐
Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**
Time Range **N/A**
Notes _____

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**
Interpass Temp. Max. **238°C**
Preheat Maintenance **N/A**
Notes _____

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-006-1 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+ EP(SMAW)
 Amps (Range) 90-140(GTAW) + 80-130(SMAW) Volts (Range) 9-13(GTAW) + 18-24(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 6.0-10.0 Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	90-130	9-12	6.0-10.0	
2nd	GTAW	ER 70S-G	2.4	DC EN	100-140	10-13	6.0-10.0	
3 & Over	SMAW	E8016-G	3.2	AC/DCEP	80-130	18-24	6.0-10.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision**.

Prepared by: Mr. Chatchawan S
 Name

Signed

Date

Reviewed by: Mr. Saksan ch
 Name

Signed

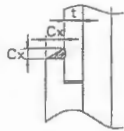
Date

Approved by: AMNUAY W
 Name

Signed

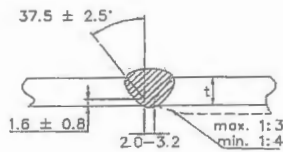
Date

ATTACHED FIGURE JOINT SKETCH



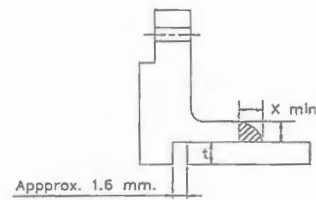
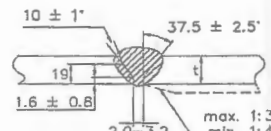
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

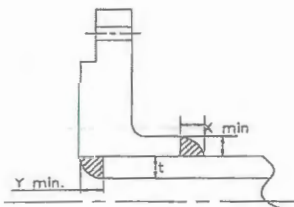
JOINT No. B1 & B2



Approx. 1.6 mm.

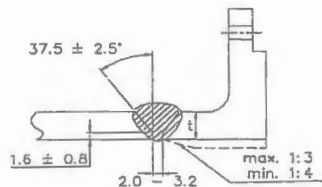
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



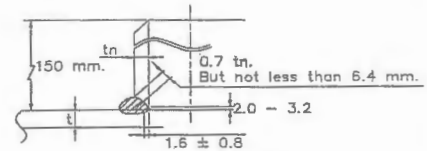
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



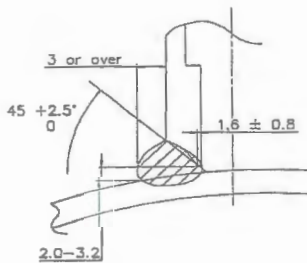
Misalignment (max.) = 1.5 mm.

JOINT No. B3

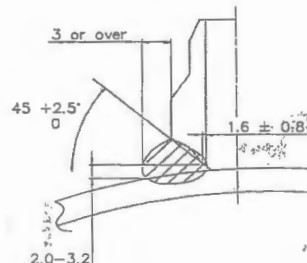


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

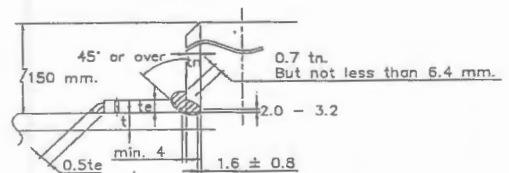
JOINT No. B4



JOINT No. C1

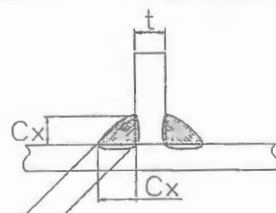


JOINT No. C2



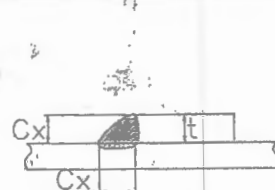
Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5



JOINT NO. D1

Theoretical throat



JOINT NO. D2

Note : C_x = Size of weld as specified
Theoretical throat = $0.707 \times C_x$

Procedure Qualification Record (PQR)

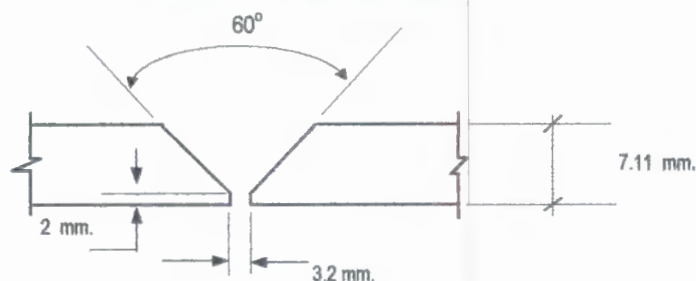
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-006-1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-006-1	Date :	13 June 2012
WPS Number :	RNC-WPS-006-1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A333	To	A333	Temperature	N/A		
Type or Grade	Gr.6	To	Gr.6	Hold Time	N/A		
P. No.	P.1	To	P.1	Others			
Thickness of Test Coupon		7.11 mm.		GAS (QW-408)			
Diameter of Test Coupon		Ø 6"					
Others							
FILLER METALS (QW-404)				Percent Composition			
	GTAW	SMAW			Gas (es)	Mixture	Flow Rate
SFA Specification	A 5.28	A 5.5		Shielding	Argon	99.95%	15 L/Min.
AWS Classification	ER70S-G	E8016-G		Trailing	-	-	-
Filler Metal F-No.	6	4		Backing	-	-	-
Weld Metal Analysis A-No.	1	1		ELECTRICAL CHARACTERISTICS (QW-409)			
Size of Filler Metal	Ø 2.4 mm.	Ø 3.2 mm.		Current	DC (GTAW)	DC (SMAW)	
Others	Kobelco TGS-1N	Kobelco NB-1		Polarity	EN	EP	
Weld Metal Thickness	-			Amperage	Voltage		
POSITION (QW-405)				10-11 for GTAW and 10-11 for SMAW			
Position of Groove	6G			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Progression of Welding (Uphill or Down)	Up Hill			Others	N/A		
Others	N/A			TECHNIQUE (QW-410)			
PREHEAT (QW-406)				Travel Speed	7 - 8 cm/min.		
Preheat Temperature	Ambient Temperature			String or Weave Bead	Both		
Interpass Temperature (max.)	222° C			Oscillation	N/A		
Others	N/A			Multipass or single Pass	Multiple Pass		
				Multipass or Single Electrodes	Multiple Pass		
				Others	N/A		

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- **RNC-PQR-006-1**

Tensile Test Report No.TS-12-06-105

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2366/12TR1	19.49	7.14	139.159	63.508	456.370	OUT OF WELD
2366/12 TR2	19.58	7.26	142.151	64.881	456.422	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-12-06-103

Type and Figure Number	Type of Bend	Indication	Results
2366/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
2366/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
2366/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
2366/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR - 005
Hardness Testing Report No. HN-12-06-080
Macrostructure Report No. MA-12-06-102
Impact Testing Report No.IM-12-06-139

Welders Name Mr.Decha T. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

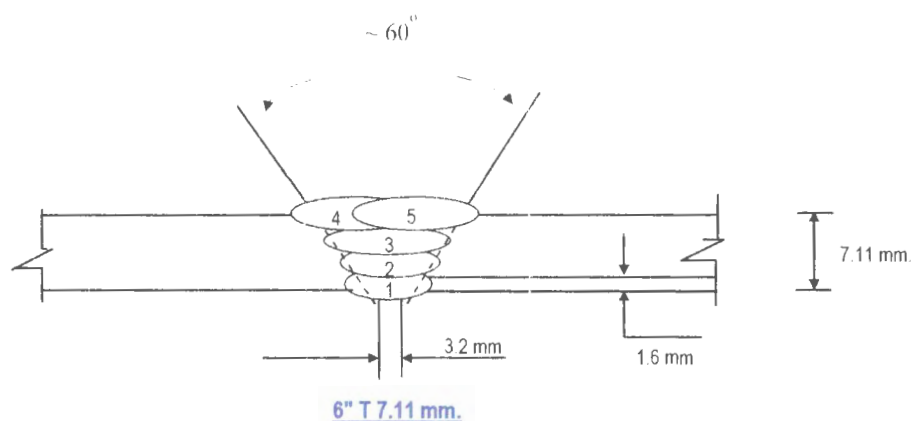
We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : 
Date : 29-06-25

Inspected by : Mr.Nattawut W.
Certified by : Mr.Chalermkiet R.
Date : 30 Jun 2012

	WELDING PROCEDURE	PQR No. : RNC-PQR-006-1
	QUALIFICATION RECORDS	WPS No. : RNC-WPS-006-1
		THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for RNC-PQR-006-1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	111	10	DCEN		-	7	9.51	15	-
2	GTAW	ER70S-G	2.4	127	11	DCEN	-	-	7	11.97	15	-
3	SMAW	E8016-G	3.2	102	23	DCEP	-	-	8	17.59	-	-
4	SMAW	E8016-G	3.2	108	23	DCEP	-	-	8	18.63	-	-
5	SMAW	E8016-G	3.2	117	23	DCEP	-	-	7	23.06	-	-



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Radiographic Examination Report

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

Page : 1 Of 1

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Mechanical Test Report



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210 Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th



21713

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-105
Revision No. 0

Test Product	PQR No. RNC-PQR-006-1	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 6" Sch 40 (7.11 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2366/12 TR1	N/A	7.14	19.49	139.159	N/A	N/A	N/A	63.508	456.370	N/A	OUT OF WELD
2	2366/12 TR2	N/A	7.26	19.58	142.151	N/A	N/A	N/A	64.881	456.422	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-006-1

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-70S-G+E8016-G/TGS-1N+NB 1

Reported by :
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date 29 / 06 / 12



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date 29 / 06 / 12
LPN PLATE MILL

- Notes**
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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

21721

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-103

Revision No. 0

Test Product	PQR No. RNC-PQR-006-1	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" Sch 40 (7.11 mm Thk.)	Diameter of Mandrel	30 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2366/12 BF1	Transverse face bend	7.24x38.09x163.08 mm	No Open Discontinuity	N/A	
2	2366/12 BF2	Transverse face bend	7.21x38.13x162.69 mm	No Open Discontinuity	N/A	
3	2366/12 BR1	Transverse root bend	7.27x38.14x163.75 mm	No Open Discontinuity	N/A	
4	2366/12 BR2	Transverse root bend	7.20x38.11x164.21 mm	No Open Discontinuity	N/A	

Additional details : WPS No. RNC-WPS-006-1

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-70S-G+E8016-G/TGS-1N+NB 1

Reported by : 

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12

Approved by : 

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12



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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



21729

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-139

Revision No. 0

Test Product	PQR No. RNC-PQR-006-1	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" Sch 40 (7.11 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2366/12-1	-20°C	5.015	10.017	54.70	137.94	132.84	Base Metal	-
2	2366/12-2	-20°C	5.011	10.014	54.73	126.54			
3	2366/12-3	-20°C	5.015	10.011	54.70	134.05			
4	2366/12-1	-20°C	5.018	10.014	54.96	114.68	130.67	Weld Metal	-
5	2366/12-2	-20°C	5.012	10.016	54.93	139.89			
6	2366/12-3	-20°C	5.010	10.015	54.99	137.45			
7	2366/12-1	-20°C	5.015	10.012	54.94	147.36	152.37	HAZ	-
8	2366/12-2	-20°C	5.013	10.016	54.93	152.28			
9	2366/12-3	-20°C	5.008	10.017	54.97	157.47			

Additional details : WPS No. RNC-WPS-006-1

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-70S-G+E8016-G/TGS-1N+NB 1

Reported by : 

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by : 

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

11065

Macrostructure Analysis Report

Page 1 of 1

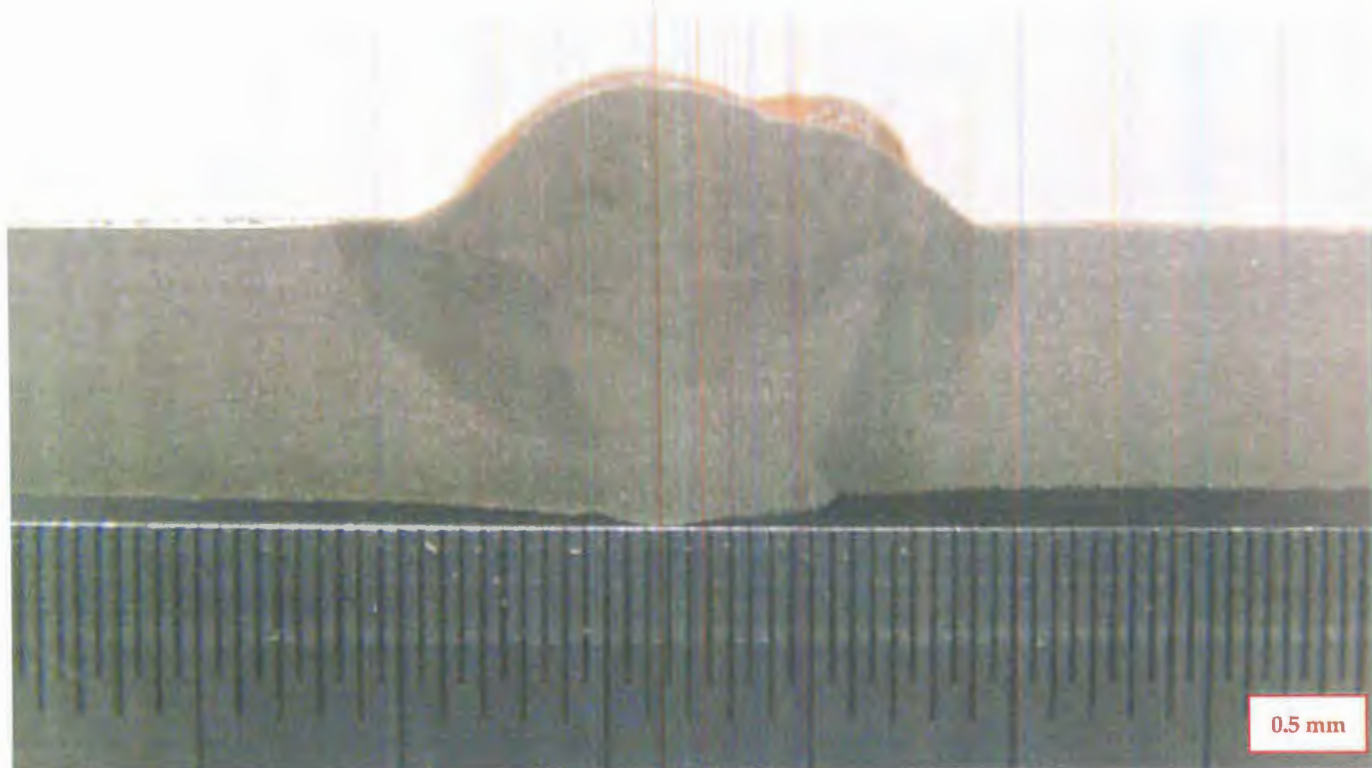
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-102

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-006-1	Material Characterization	Pipe
Test No.	2366/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A333Gr.6	Temperature	28 °C
Dimension	Ø6"Sch40(7.11 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW+SMAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks
 Filler/Electrode Spec't : ER70S-G+E7016-G TGS-IN+NB 1

Reported by : N. Keerati Sa-ngoen(Mr.Keerati Sa-ngoen)
Laboratory Section Manager

Date : 29/06/12

Approved by : N. Pavaret Preedawiphat(Mr.Pavaret Preedawiphat)
Deputy Managing Director

Date : 29/06/12

LPN PLATE MILL**Notes**

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited

12407

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-080

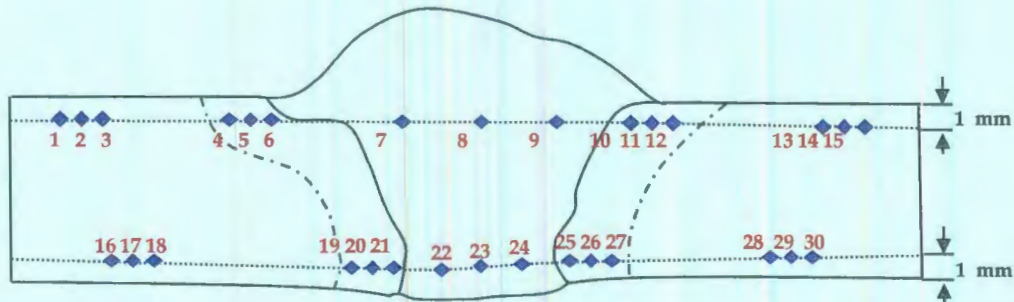
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-006-1	Material Characterization	Pipe
Test No.	2366/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333Gr.6	Equipment	Mitutoyo, HV-115
Dimension	Ø6"Sch40(7.11 mm Thk.)	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW+SMAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

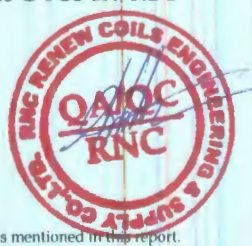
Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	129.0	16	BASE	133.7	*****		
2	BASE	128.0	17	BASE	132.2			
3	BASE	128.3	18	BASE	133.6			
4	HAZ	144.3	19	HAZ	143.0			
5	HAZ	152.9	20	HAZ	147.9			
6	HAZ	170.6	21	HAZ	156.8			
7	WELD	211.3	22	WELD	150.8			
8	WELD	184.5	23	WELD	156.8			
9	WELD	195.4	24	WELD	157.1			
10	HAZ	177.8	25	HAZ	157.5			
11	HAZ	161.4	26	HAZ	151.1			
12	HAZ	154.5	27	HAZ	151.5			
13	BASE	130.8	28	BASE	131.6			
14	BASE	128.8	29	BASE	132.6			
15	BASE	129.8	30	BASE	131.3			

Additional details : Filler/Electrode Spec't : ER70S-G+E8016-G TGS-1N+NB 1

Reported by :
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date : 21 / 06 / 12



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 21 / 06 / 12

Notes

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2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T TABMA A, MUANG RAYONG 21000

TEL : (038) 663 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Consumable Certificate

TKW110514 KOBELCO

INSPECTION CERTIFICATE
(検査証明書)PURCHASER
需要家COVERED ELECTRODE
被覆アーク溶接棒CERTIFICATE No.
証明書番号: 120112100
DATE OF ISSUE
発行日: 2011.08.18

TRADE DESIGNATION 品名(型番)	DIMENSION 寸法(mm)	MFG.No 製造番号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類
NB-1	(3.2) mm 1/8 inch	A171	AMS A5.5 E8016-6 ASME SFA-5.5 E8016-6

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	NI							
DEPOSITED METAL 溶着金属	0.06	0.62	1.04	0.009	0.003	1.69							
ELEMENTS 成分													
DEPOSITED METAL 溶着金属													

TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能				HARDNESS PROPERTIES 硬さ	
YIELD POINT	YIELD STRENGTH AT 0.2% OFFSET	TENSILE STRENGTH	ELONGATION	TEST TEMP.	ABSORBED ENERGY 吸収エネルギー				
降伏点	0.2% 耐力	引張強さ	伸び	試験温度	AVG. 平均	EACH			
(-) N/mm ²	(≥460) N/mm ²	(≥550) N/mm ²	≥19 %	(-60) °C	(≥27) J	(≥21)			
- ksi	≥67 ksi	≥80 ksi		-76 °F	≥20 ft-lbf	≥15			

WELDING CONDITIONS 溶接条件				POSTWELD HEAT TREATMENT 溶接後熱処理	
TYPE OF CURRENT 電流の種類	AC	SHIELDING GAS シールドガス	-		
AMPERAGE 溶接電流	115 A			(620) °C × 1 h	
ARC VOLTAGE アーク電圧	21 V			1148 °F × 1 h	

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.

上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。

◆KOBELCO STEEL, LTD.
WELDING BUSINESS IBARAKI PLANT
株式会社 神戸製鋼所 溶接事業部門 茨木 工場

N. Cerachi

APPROVED BY CHIEF INSPECTOR
主任検査員 承認済

TKW110801 HOBELCO

INSPECTION CERTIFICATE
(検査証明書)PURCHASER
需要家WELDING ROD OR WIRE
溶加棒又はワイヤ

CERTIFICATE No.

証明書番号: 520113299

DATE OF ISSUE

発行日: 2011.10.28

TRADE DESIGNATION 品名(型番)	DIMENSION 寸法(mm)	MFG.No 製造番号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類
TG-S1N	(2.4) mm 3/32 inch	GZ00820	AWS A5.28 ER70S-G ASME SFA-5.28 ER70S-G

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	CU	NI	MO					
ROD OR WIRE 棒又はワイヤ	0.06	0.31	1.04	0.006	0.009	0.11	0.80	0.14					
ELEMENTS 成分									FS = FERRITE % (SCHAFFLER DIAGRAM) FN = FERRITE NUMBER (DELONG DIAGRAM) FNM = FERRITE NUMBER (WRC)				
ROD OR WIRE 棒又はワイヤ													

TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能				HARDNESS PROPERTIES 硬さ	
YIELD POINT 降伏点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引張強さ	ELONGATION 伸び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸収エネルギー				
(-) N/mm ² ksi	(≥360) N/mm ² ksi	(≥480) N/mm ² ksi	≥24 %	(-60) °C -73 °F	AVG. 平均 (≥27) J	EACH (≥21)			
	≥52	≥70			≥20 ft-lb	≥15			

WELDING CONDITIONS 溶接条件				POSTWELD HEAT TREATMENT 溶接後熱処理	
TYPE OF CURRENT 電流の種類	DOEN	SHIELDING GAS シールドガス	AR 100%		
AMPERAGE 溶接電流	220 A			(-) °C × - h	
ARC VOLTAGE アーク電圧	14 V			- °F × - h	

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.

上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。

◆KOBELCO STEEL, LTD.

WELDING BUSINESS

株式会社 神戸製鋼所 溶接事業部門

FUKUCHIYAMA PLANT

福知山工場

APPROVED BY CHIEF INSPECTOR

主任検査員 承認済



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T IABMA, A MUANG RAYONG 21000

TEL : (038) 663 476 FAX : (038) 633 477 MOBILE : (086) 340-9978

Material Certificate

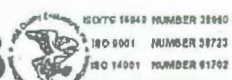


VALLOUREC & MANNESMANN TUBES

V & M do BRASIL S.A.

BARREIRO PLANT - Belo Horizonte - MG - Brazil

CEP: 30161-970 - PO BOX: 1453



INSPECTION CERTIFICATE
(According to DIN EN 10204 3.1)

Nº.: 0030021384

Sheet: 1 / 2

Customer: HUPSTEEL LIMITED

Customer Order: AR2363-0311

Inspection: V & M do BRASIL S.A.

Country: Singapore

Work Order: 294024 / 000010

Material Number: 384831

Customer Material Number:

DIMENSION: 168.30mm X 7.11mm SCHEDULE: 040 # GRADE: GR 6
METALURGICAL STANDARD: ASTM A 333 01.04.2010 # DIMENSIONAL STANDARD: ASTM A 333 01.04.2010
IN ACCORDANCE ALSO TO THE STANDARDS: ASME SA-333 / 2010 #
PRODUCT: SEAMLESS STEEL PIPE, HOT FINISHED, BEVELED ENDS 37.5 DEG. # NORMALIZED # SURFACE PROTECTION: LACQUER #
TOLERANCES: OUTSIDE DIAMETER: - 0.800mm / + 1.600mm WALL THICKNESS: - 0.889mm / + 1.600mm
CUT LENGTH: FIXED 6000.00mm
STANDARD MARKING: VMB ASTM A/ASME SA-333 6 S HF SCH 040 LT -50 C HEAT "HEAT"
SHIPPING MARKING: HUPSTEEL * P.O NO.: 5001122* SINGAPORE * NACE MR 01.75

Heat N.	Inspection Lot	Pieces
109285	030001927413	284
109286 /	030001927414	100
Total		384

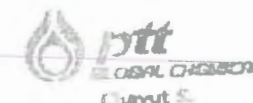
THE PRODUCT IS SATISFACTORY IN THE FOLLOWING TESTS/INSPECTIONS: DIMENSIONAL # VISUAL # FLATTENING TEST # ELECTROMAGNETIC TEST ASTM E 570 # HYDROSTATIC TEST: 12,0 MPA 5 s #

Chemical Composition(%)		CEq according: C+Mn/6+(Cr+Mo+V)/5 +(Ni+Cu)/15										DI according:		Unit:	PCM:	
		C	Mn	P	S	Si	Ni	Cr	Mo	Al	Cu	V	Nb	Ti	CEq	
Required:	Min	0.29				0.10										
	Max	0.20	1.06	0.025	0.010		0.25	0.20	0.06		0.350				0.410	
Heat N./Inspection Lot																
109285	030001927413	0.10	0.91	0.012	0.004	0.17	0.01	0.04	0.01	0.023	0.010	0.000	0.019	0.001	0.263	
	Check1	0.11	0.90	0.012	0.004	0.18	0.01	0.04	0.01	0.026	0.006	0.000	0.018	0.001	0.269	
	Check2	0.11	0.93	0.013	0.006	0.18	0.01	0.05	0.01	0.023	0.006	0.003	0.020	0.001	0.278	
109286	030001927414	0.10	0.94	0.014	0.005	0.16	0.01	0.05	0.01	0.025	0.000	0.000	0.019	0.001	0.269	
	Check1	0.11	0.94	0.015	0.005	0.17	0.01	0.05	0.01	0.028	0.003	0.000	0.020	0.001	0.276	
	Check2	0.11	0.95	0.015	0.004	0.17	0.01	0.05	0.01	0.025	0.004	0.000	0.019	0.001	0.276	

Mechanical Tests S.Direction: Longitudinal Temperature: Room Temperature Gage Length: L0=2"

		Area(MM2)	YS(MPA)	TS(MPA)	E(%)
Required:	Min		240	415	28
	Max				
Heat N./Inspection Lot					
109285	030001927413	177.4	331	441	40
109286	030001927414	192.2	329	437	42

YS-Yield Strength ; TS-Tensile Strength ; E-Elongation



Inspection QA/QC Division
Integrity and Reliability

CERTIFIED TRUE COPY

BY DELTAMETAL(THAILAND) LIMITED

P.O.NO. 45015361

ITEM NO. 2 09105

Hardness

		HV
Required:	Min	
	Max	246
Heat N./Inspection Lot		
109285	030001927413	145
109286	030001927414	146

Impact According: CHARPY 10X55X5 V NOTCH Direction: Longitudinal Temperature: -50 °C

		AE1 (J)	AE2 (J)	AE3 (J)	AE Aver. (J)
Required:	Min	14	14	14	17,0
	Max				
Heat N./Inspection Lot					
109285	030001927413	98	110	114	107,3
109286	030001927414	122	120	116	119,3

AE-Absorbed Energy

Remarks:

- MATERIAL IN ACCORDANCE TO ITEM A.2 NACE MR 0175/ISO 15156-2/2003.

- FINAL HOT ROLLING OPERATION FINISHED ABOVE 850°C AND COOLED IN STILL AIR

We would like to confirm that for the above stated item, the pipes were stencilled with customer name as #HUPSTEL# instead of #HUPSTEEL# due to the problem in our computer system.

pte
GLOBAL CHARTER
Chayut S.
Inspection QA/QC Division
Integrity and Reliability



We hereby certify that this product has been manufactured and examined in accordance with all requirements of the standards and specifications and all the results are found to be satisfactory. This testimonial and certificate respectively is recorded by a computer system and is valid without signature. Alteration or use for others products are regarded as falsification of documents and will be subject to criminal jurisdiction.

QUALITY CONTROL DEPARTMENT

FAX: (55-31) 3328-2587

e-mail: CONT.QUALIDADE@VMTUBES.COM.BR

DATE
30.05.2011

LUIZ ALEXANDRE COSTA - CREA/MG 60.548/D
TECHNICAL RESPONSIBLE

Eco Tubes: The tubes from V&M do BRASIL are manufactured with steel which uses charcoal as a source of energy in its production. This coal comes from more than 100.000 ha of forest planted by V&M Florestal. With the acquisition of 67,9 ton(s) of steel tubes from V & M do BRASIL, your company contributed to the reduction of the greenhouse effect, avoiding the accumulation of 122,2 ton(s) of Carbon Dioxide CO2 in the atmosphere.

CERTIFIED TRUE COPY
BY DELTAMETAL (THAILAND) LIMITED
P.O. NO. 49015261
ITEM NO. 2000